MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – D)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The equation of a line parallel to y-axis and at a distance of 5 units to the right of y-axis is : [1]

(A) $x = 5$ (B) $x = -5$ (C) $y = 5$ (D) $y = -5$

Q2. The equation of a line parallel to the x-axis and at a distance of 3 units below x-axis is : [1]

(A) $x = 3$ (B) $x = -3$ (C) $y = -3$ (D) $y = 3$

Q3. The line represented by the equation $x - y = 0$ is : [1]

(A) parallel to x-axis (B) parallel to y-axis (C) passing through the origin (D) passing through the point $(3, 2)$

Q4. The distance between the points $(4\cos\theta + 3\sin\theta, 0)$ and $(0, 4\sin\theta - 3\cos\theta)$ is : [1]

(A) 25 (B) 7 (C) 5 (D) $\sqrt{7}$

Q5. 5 trees Banyan, Silver Oak, Orange, Amla and Mango were planted in the garden as shown in the figure. If 1 unit = 1 m in the figure, which of the following trees are at a distance of 4 metres ? *(From diagram: Orange $(0, 7)$, Silver Oak $(3, 4)$, Banyan $(-3, 4)$, Mango $(2, 0)$, Amla $(-2, 0)$)* [1]

(A) Amla and Mango (B) Mango and Silver Oak (C) Silver Oak and Orange (D) Orange and Mango

SECTION B: Short Answer Type Questions (10 Marks)

Q6. The coordinates of the end points of the line segment AB are $A(-2, -2)$ and $B(2, -4)$. P is the point on AB such that $BP = \frac{4}{7}AB$. Find the coordinates of point P. [2]

Q7. In the given figure, point D divides the side BC of ΔABC in the ratio 1 : 2. Find length AD. *(From given figure: $A(1, 5)$, $B(-2, 1)$, $C(4, 2)$).* [2]

SECTION B (Continued): Short Answer Type Questions (Questions 8 to 10)

Q8. Find the coordinates of the point which divides the line segment joining $(-1, 3)$ and $(4, -7)$ internally in the ratio $3 : 4$. **[2]**

Q9. Find the coordinates of the point C which lies on the line AB produced such that $AC = 2BC$, where coordinates of points A and B are $(-1, 7)$ and $(4, -3)$ respectively. **[2]**

Q10. The coordinates of the centre of a circle are $(2a, a - 7)$. Find the value(s) of 'a' if the circle passes through the point $(11, -9)$ and has diameter $10\sqrt{2}$ units. **[2]**

SECTION C: Long Answer & Competency Based Questions (15 Marks)

Q11. Find the ratio in which $P(4, m)$ divides the line segment joining the points $A(2, 3)$ and $B(6, -3)$. Hence find m. **[5]**

Q12. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts. **[5]**

Q13. If the points $A(k + 1, 1), B(2k, 3), C(2k - 1, 2k - 1)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$, find the value of k. **[5]**

MATHS MANTHAN
TUITION CLASSES

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★